

PROJECT DATA SHEET



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JOB/PROJECT : NOSS ON DART MARINA, DARTMOUTH

JOB/PROJECT NO : C8829 SHEET NO : COVER

GENERAL :

CLIENT : PREMIER MARINAS

ARCHITECT : N/A

AUTHORITY : TORBAY COUNCIL

JOB/PROJECT DESCRIPTION : PILING PLATFORM

PROPOSED USE: PILING PLATFORM FOR GEAX EK60L

OVERALL SIZE : AS CALCS

No. OF STOREYS : N/A

TYPE OF SOIL INVESTIGATION : TRIAL PITS, BOREHOLES

BEARING STRATA : WEAK MUDSTONE

TYPE OF FOUNDATION : N/A

MAXIMUM SAFE BEARING PRESSURE

N/A kN/m²

STRUCTURAL MATERIALS USED :

STANDARDS AND CODES USED IN DESIGN :

General actions: Densities, self weight & imposed loads	BS EN 1991-1-1	☒	Steelwork	BS EN 1993 (EC3)	☐
			Timber	BS EN 1995 (EC5)	☐
General actions: Actions on structures exposed to fire	BS EN 1991-1-2	☐	Masonry	BS EN 1996 (EC6)	☐
Snow Loads	BS EN 1991-1-3	☐	Others	TWF 2019:02 DESIGN OF GRANULAR WORKING PLATFORMS FOR CONSTRUCTION PLANT.	☒
Wind Loads	BS EN 1991-1-4	☐			☐
Concrete	BS EN 1992 (EC2)	☐			☐

AUTHORISATION OF CALCULATION SHEETS AND BENDING SCHEDULES :

CALCULATION SHEET NOS.	CHECKED BY	DATE
Sheet nos: G01 - G32 inclusive	<i>D.C. HEARD</i>	<i>15/12/23</i>

 CASE Consultants Civil And Structural Engineers Moose Hall Barewell Rd Torquay, TQ1 4PA	Project Noss On Dart Marina, Dartmouth		Job Ref. C8829	
	Section GEAX EK60L Piling Rig		Sheet no./rev. G 1	
	Calc. by D.P.H	Date DEC 23	Chk'd by	Date
			App'd by	Date

Description

These calculations cover the design of the piling platform at Noss on Dart Marina, Dartmouth, adjacent to the proposed king post retaining wall. This check is to assess the capacity for the GEAX EK60L Piling Rig.

The maximum safe ground bearing pressure and soil parameters have been taken from a SOILS Ltd investigation document No.20652/R1 and accompanying online resources.

Information for the piling rigs has been taken from Sheppard Piling working platform certificate for the GEAX EK60L rig, giving two load cases.

Equipment to be used on site.	GEAX EK60L CFA piling rig
Maximum plant loading	BRE Load Case 1: q1k = 123kPa, L1 = 2.31m, W1 = 0.5m BRE Load Case 2: q2k = 143kPa, L2 = 1.95m, W2 = 0.5m

These two different cases will be considered with the two different types of soil identified in the SOILS Ltd report, made ground (MG) comprised of slightly silty very clayey GRAVEL and Bovisand Formation (BOV) consisting of slaty MUDSTONE.

Contents

Page P2 - P14 inc – Made Ground (slightly silty very clayey GRAVEL)
Page P15 - P27 inc – Bovisand Formation (slaty MUDSTONE)
Page P28 – P31 inc – Tensar Geogrid Analysis
Page P32 – Conclusion



Made ground

ULS check - LC1 and clayey GRAVEL

1. VARIABLE ACTIONS

applied load breadth:

$$B := 0.5 \text{ m}$$

applied load length:

$$L := 2.31 \text{ m}$$

applied load area:

$$A := B \cdot L = 1.155 \text{ m}^2$$

characteristic applied GBP:

$$q_k := 123 \text{ kPa}$$

Partial factors on variable actions:

$$\gamma_Q := 1.3$$

design applied ground bearing pressure:

$$q_d := \gamma_Q \cdot q_k = 159.9 \text{ kPa}$$

Design applied load:

$$Q_d := q_d \cdot A = 184.7 \text{ kN}$$

2. SUB-GRADE PARAMETERS

Subgrade material characteristic angle of internal friction:

$$\varphi_{sk} := 28^\circ$$

Partial factor on sub-grade strength:

$$\gamma_\varphi := 1.25$$

Sub-grade material design angle of internal friction:

$$\varphi_{sd} := \text{atan} \left(\frac{\tan(\varphi_{sk})}{\gamma_\varphi} \right) = 23.04 \text{ deg}$$

$$\varphi_{sd} := 23.04$$

sub-grade material density:

$$\gamma_s := 18 \frac{\text{kN}}{\text{m}^3}$$

depth of ground water below formation:

$$z_\gamma := 0.82 \text{ m}$$

3. SUB-GRADE BEARING RESISTANCE WITHOUT PLATFORM

Bearing capacity factor for gravity term:

$$N_{\gamma d} := 0.1054 \cdot e^{0.168 \cdot \varphi_{sd}} = 5.06$$

design shape factor for gravity term:

$$s_{\gamma d} := 1 - \left(0.4 \cdot \frac{B}{L} \right) = 0.91$$

design bearing resistance:

$$V_{Rd} := 0.5 \cdot \gamma_s \cdot B \cdot N_{\gamma d} \cdot s_{\gamma d} \cdot A = 24.01 \text{ kN}$$

UTILISATION:

$$\frac{Q_d}{V_{Rd}} = 7.69$$

> 1.0 therefore FAIL &
Platform required



4. PLATFORM PARAMETERS

select platform material based on presumed bearing capacity (table 3):

platform material characteristic angle of internal friction:

$$\varphi_{pk} := 40^\circ$$

partial factor on platform material strength:

$$\gamma_\varphi := 1.25$$

platform material design angle of internal friction:

$$\varphi_{pd} := \text{atan} \left(\frac{\tan(\varphi_{pk})}{\gamma_\varphi} \right) = 33.87 \text{ deg}$$

platform material density:

$$\gamma_p := 20 \frac{\text{kN}}{\text{m}^3}$$

5.1 EFFECTIVE AREA, DIMENSIONS AND LOAD SPREAD ANGLE

PLATFORM THICKNESS:

$$D := 0.45 \text{ m}$$

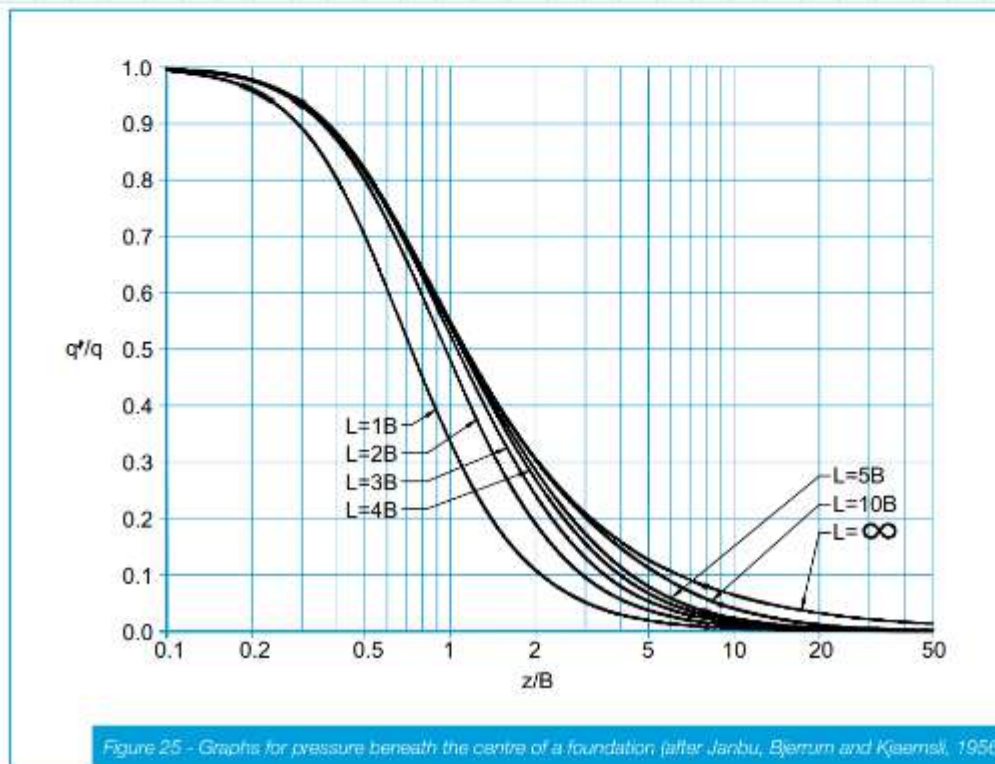
find ratio of applied pressure to effective pressure at formation (figure 25)

ratio of applied load breadth to platform depth:

$$\frac{D}{B} = 0.9$$

ratio of applied load length to breadth:

$$\frac{L}{B} = 4.62$$





ratio of applied pressure to effective pressure
(from chart):

$$\rho_q := \frac{q_d}{q_d} \quad \rho_q := 0.58$$

design effective pressure:

$$q_d' := \rho_q \cdot q_d = 92.74 \text{ kPa}$$

design effective area:

$$A' := \frac{Q_d}{q_d'} = 1.99 \text{ m}^2$$

final load spread width and angle solving with quadratic equation

quadratic factors:

$$\begin{aligned} a &:= 1 \\ b &:= L + B = 2.81 \text{ m} \\ c &:= A - A' = -0.84 \text{ m}^2 \end{aligned}$$

quadratic solution:

$$x := \frac{\sqrt{(b^2 - 4 \cdot a \cdot c)} - b}{2 \cdot a} = 0.27 \text{ m}$$

load spread width:

$$b' := \frac{x}{2} = 0.14 \text{ m}$$

load spread angle:

$$\beta := \text{atan}\left(\frac{b'}{D}\right) = 16.78 \text{ deg}$$

$$\beta := 16.78$$

maximum load spread width for $\beta = 26.6$

$$b'_{max} := \frac{D}{2} = 0.225 \text{ m}$$

$b' < b'_{max}$ therefore use b'

effective breadth:

$$B' := \min(B + 2 \cdot b', B + 2 \cdot b'_{max}) = 0.77 \text{ m}$$

effective length:

$$L' := \min(L + 2 \cdot b', L + 2 \cdot b'_{max}) = 2.58 \text{ m}$$

5.2 EFFECTIVE AREA OF PUNCHING SHEAR

find design punching shear angle (figure 27)

platform material characteristic angle of internal friction:

$$\varphi_{pk} = 40 \text{ deg}$$

Subgrade material characteristic angle of internal friction:

$$\varphi_{sk} = 28 \text{ deg}$$

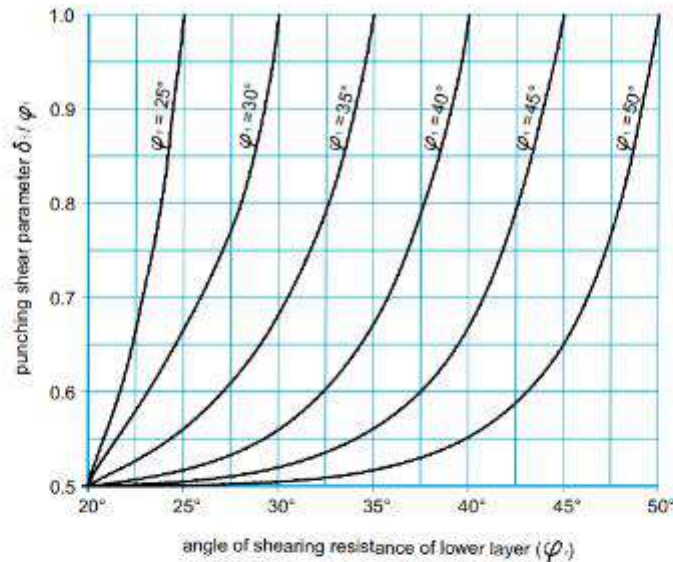


Figure 27 – δ/ϕ for granular formation (after Hanna, 1981)

nominal punching shear parameter (from chart):

$$\rho_{\delta} := \frac{\delta_{pk}}{\varphi_{pk}} \quad \rho_{\delta} := 0.53$$

design punching shear angle:

$$\delta_{pd} := \rho_{\delta} \cdot \varphi_{pd} = 17.95 \text{ deg}$$

5.3 LATERAL LOADS IN PLATFORM

coefficient of active lateral earth pressure for platform:

$$K_{apd} := \left(\frac{\sin(90^\circ - \varphi_{pd})}{\sqrt{\sin(90^\circ + \delta_{pd})} + \sqrt{(\sin(\varphi_{pd} + \delta_{pd}) \cdot \sin(\varphi_{pd}))}} \right)^2 = 0.26$$

$$K_{ppd} := \left(\frac{\sin(90^\circ + \varphi_{pd})}{\sqrt{\sin(90^\circ - \delta_{pd})} - \sqrt{(\sin(\varphi_{pd} + \delta_{pd}) \cdot \sin(\varphi_{pd}))}} \right)^2 = 7.02$$

find ratio of applied pressure to effective pressure at mid-depth of platform (figure 25)

mid-point depth of platform

$$D_{mid} := \frac{D}{2} = 0.225 \text{ m}$$

mid-point depth to breadth ratio:

$$\frac{D_{mid}}{B} = 0.45$$

ratio of applied load length to breadth:

$$\frac{L}{B} = 4.62$$

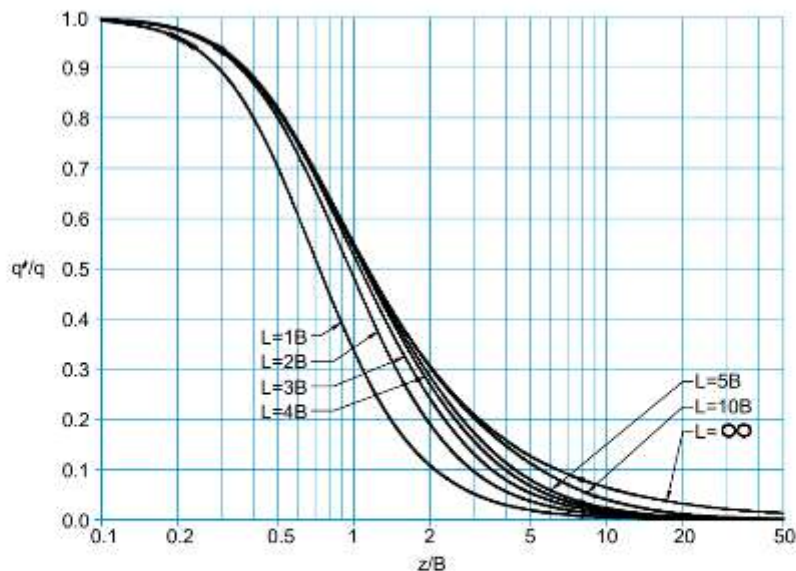


Figure 25 - Graphs for pressure beneath the centre of a foundation (after Janbu, Høegem and Kjaernvik, 1956)

ratio of applied pressure to effective pressure (from chart) $\rho_q := \frac{q_{avd}}{q_d}$ $\rho_q := 0.85$

average vertical pressure due to load:

$$q_{avd}' := \rho_q \cdot q_d = 135.92 \text{ kPa}$$

active lateral load in fill (per lin m):

$$P_{apd} := K_{apd} \cdot \left(q_{avd}' + \gamma_p \cdot \frac{D}{2} \right) \cdot D = 16.25 \frac{\text{kN}}{\text{m}}$$

passive lateral load in fill (per lin m)

$$P_{ppd} := K_{ppd} \cdot \gamma_p \cdot \frac{D^2}{2} = 14.21 \frac{\text{kN}}{\text{m}}$$

5.4 HORIZONTAL AND VERTICAL LOADS ON SUB-GRADE

horizontal load on sub-grade (per lin m)

$$F_{Hs} := P_{apd} - P_{ppd} = 2.04 \frac{\text{kN}}{\text{m}}$$

vertical load on sub-grade (per lin m)

$$F_{Vs} := \frac{(q_d \cdot B + \gamma_p \cdot D \cdot B)}{2} = 43.45 \frac{\text{kN}}{\text{m}}$$

5.5 TOTAL VERTICAL LOAD EFFECT ON SUB-GRADE

characteristic permanent action due to platform self-weight $G_{pk} := \gamma_p \cdot D \cdot B' \cdot L' = 17.92 \text{ kN}$

partial factor for permanent actions:

$$\gamma_G := 1.0$$

design permanent action due to platform self-weight

$$G_{pd} := \gamma_G \cdot G_{pk} = 17.92 \text{ kN}$$

total design vertical action

$$V_{Ed} := G_{pd} + Q_d = 202.61 \text{ kN}$$



5.6 SUB-GRADE BEARING RESISTANCE WITH PLATFORM

bearing capacity factor for gravity term

$$N_{\gamma d} = 5.06$$

Sub-grade material design angle of internal friction (see Section 2):

$$\varphi_{sd} = 23.04 \quad \varphi_{sd} := 23.04^\circ$$

bearing capacity factor for overburden term

$$N_{qd} := e^{\pi \cdot \tan(\varphi_{sd})} \cdot \left(\tan \left(45^\circ + \frac{\varphi_{sd}}{2} \right) \right)^2 = 8.7$$

shape factor for gravity term:

$$s_{\gamma d} := 1 - \left(0.4 \cdot \frac{B'}{L'} \right) = 0.88$$

Shape factor for overburden term:

$$s_{qd} := 1 + \left(\tan(\varphi_{sd}) \cdot \frac{B'}{L'} \right) = 1.13$$

inclination factor exponent:

$$m := \frac{\left(2 + \left(\frac{B'}{L'} \right) \right)}{1 + \left(\frac{B'}{L'} \right)} = 1.77$$

inclination factor for gravity term:

$$i_{\gamma d} := \min \left(1, \left(1 - \frac{F_{Hs}}{F_{Vs}} \right)^{m+1} \right) = 0.88$$

inclination factor for overburden term:

$$i_{qd} := \min \left(1, \left(1 - \frac{F_{Hs}}{F_{Vs}} \right)^m \right) = 0.92$$

depth factor for gravity term;

$$d_{\gamma d} := 1$$

depth factor for overburden term:

$$d_{qd} := 1 + 2 \cdot \tan(\varphi_{sd}) \cdot (1 - \sin(\varphi_{sd}))^2 \cdot \operatorname{atan} \left(\frac{D}{B'} \right) = 1.17$$

groundwater factor for gravity term:

$$w_{\gamma d} := \min \left(1, \max \left(0.5, 0.5 \cdot \left(1 + \frac{z_{\gamma}}{B'} \right) \right) \right) = 1$$

depth of groundwater below top of platform:

$$z_q := z_{\gamma} + D = 1.27 \text{ m}$$

groundwater factor for surcharge term:

$$w_{qd} := \min \left(1, \max \left(0.5, 0.5 \cdot \left(1 + \frac{z_q}{D} \right) \right) \right) = 1$$

bearing resistance (overburden)

$$W_{Rd1} := 0.5 \cdot \gamma_s \cdot B' \cdot N_{\gamma d} \cdot s_{\gamma d} \cdot i_{\gamma d} \cdot d_{\gamma d} \cdot w_{\gamma d} = 27.059 \text{ kPa}$$

bearing resistance (self-weight)

$$W_{Rd2} := \gamma_p \cdot D \cdot N_{qd} \cdot s_{qd} \cdot i_{qd} \cdot d_{qd} \cdot w_{qd} = 94.507 \text{ kPa}$$

total

$$W_{Rd} := W_{Rd1} + W_{Rd2} = 121.566 \text{ kPa}$$

$$L' \cdot B' = 1.991 \text{ m}^2$$

bearing resistance

$$V_{Rd} := W_{Rd} \cdot B' \cdot L' = 242.084 \text{ kN}$$

UTILISATION

$$\frac{V_{Ed}}{V_{Rd}} = 0.837$$



ULS check - LC2 and clayey GRAVEL

1. VARIABLE ACTIONS

applied load breadth:

$$B := 0.5 \text{ m}$$

applied load length:

$$L := 1.95 \text{ m}$$

applied load area:

$$A := B \cdot L = 0.975 \text{ m}^2$$

characteristic applied GBP:

$$q_k := 143 \text{ kPa}$$

Partial factors on variable actions:

$$\gamma_Q := 1.3$$

design applied ground bearing pressure:

$$q_d := \gamma_Q \cdot q_k = 185.9 \text{ kPa}$$

Design applied load:

$$Q_d := q_d \cdot A = 181.3 \text{ kN}$$

2. SUB-GRADE PARAMETERS

Subgrade material characteristic angle of internal friction:

$$\varphi_{sk} := 28^\circ$$

Partial factor on sub-grade strength:

$$\gamma_\varphi := 1.25$$

Sub-grade material design angle of internal friction:

$$\varphi_{sd} := \text{atan} \left(\frac{\tan(\varphi_{sk})}{\gamma_\varphi} \right) = 23.04 \text{ deg}$$

$$\varphi_{sd} := 23.04$$

sub-grade material density:

$$\gamma_s := 18 \frac{\text{kN}}{\text{m}^3}$$

depth of ground water below formation:

$$z_\gamma := 0.82 \text{ m}$$

3. SUB-GRADE BEARING RESISTANCE WITHOUT PLATFORM

Bearing capacity factor for gravity term:

$$N_{\gamma d} := 0.1054 \cdot e^{0.168 \cdot \varphi_{sd}} = 5.06$$

design shape factor for gravity term:

$$s_{\gamma d} := 1 - \left(0.4 \cdot \frac{B}{L} \right) = 0.9$$

design bearing resistance:

$$V_{Rd} := 0.5 \cdot \gamma_s \cdot B \cdot N_{\gamma d} \cdot s_{\gamma d} \cdot A = 19.91 \text{ kN}$$

UTILISATION:

$$\frac{Q_d}{V_{Rd}} = 9.1$$

> 1.0 therefore FAIL &
Platform required



4. PLATFORM PARAMETERS

select platform material based on presumed bearing capacity (table 3):

platform material characteristic angle of internal friction:

$$\varphi_{pk} := 40^\circ$$

partial factor on platform material strength:

$$\gamma_\varphi := 1.25$$

platform material design angle of internal friction:

$$\varphi_{pd} := \operatorname{atan}\left(\frac{\tan(\varphi_{pk})}{\gamma_\varphi}\right) = 33.87 \text{ deg}$$

platform material density:

$$\gamma_p := 20 \frac{\text{kN}}{\text{m}^3}$$

5.1 EFFECTIVE AREA, DIMENSIONS AND LOAD SPREAD ANGLE

PLATFORM THICKNESS:

$$D := 0.45 \text{ m}$$

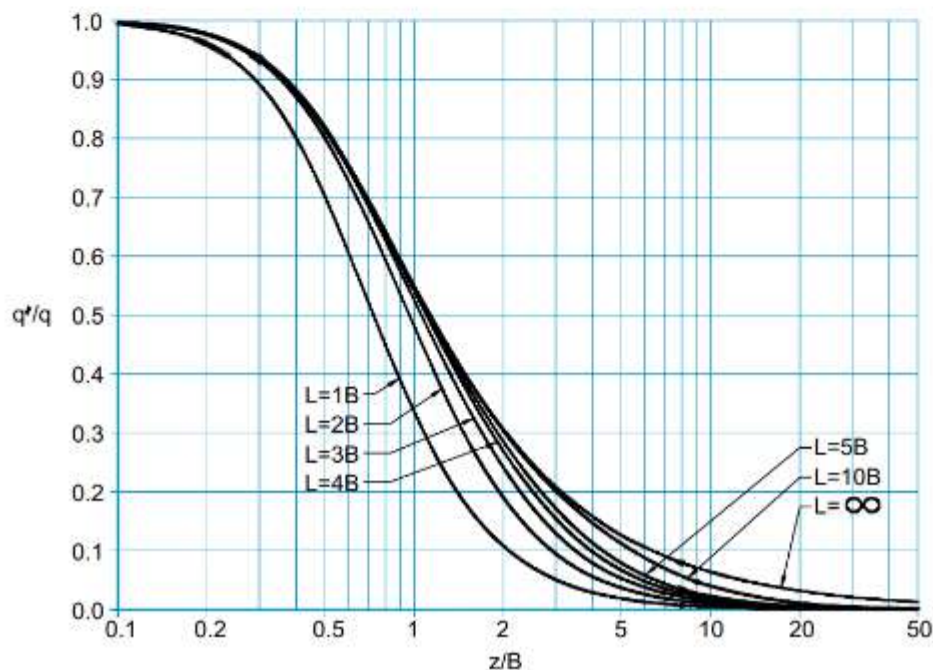
find ratio of applied pressure to effective pressure at formation (figure 25)

ratio of applied load breadth to platform depth:

$$\frac{D}{B} = 0.9$$

ratio of applied load length to breadth:

$$\frac{L}{B} = 3.9$$





ratio of applied pressure to effective pressure
(from chart):

$$\rho_q := \frac{q_d'}{q_d} \quad \rho_q := 0.57$$

design effective pressure:

$$q_d' := \rho_q \cdot q_d = 105.96 \text{ kPa}$$

design effective area:

$$A' := \frac{Q_d}{q_d'} = 1.71 \text{ m}^2$$

final load spread width and angle solving with quadratic equation

quadratic factors:

$$\begin{aligned} a &:= 1 \\ b &:= L + B = 2.45 \text{ m} \\ c &:= A - A' = -0.74 \text{ m}^2 \end{aligned}$$

quadratic solution:

$$x := \frac{\sqrt{(b^2 - 4 \cdot a \cdot c)} - b}{2 \cdot a} = 0.27 \text{ m}$$

load spread width:

$$b' := \frac{x}{2} = 0.14 \text{ m}$$

load spread angle:

$$\beta := \text{atan}\left(\frac{b'}{D}\right) = 16.72 \text{ deg}$$

$$\beta := 16.72$$

maximum load spread width for $\beta = 26.6$

$$b'_{\max} := \frac{D}{2} = 0.225 \text{ m}$$

$b' < b'_{\max}$ therefore use b'

effective breadth:

$$B' := \min(B + 2 \cdot b', B + 2 \cdot b'_{\max}) = 0.77 \text{ m}$$

effective length:

$$L' := \min(L + 2 \cdot b', L + 2 \cdot b'_{\max}) = 2.22 \text{ m}$$

5.2 EFFECTIVE AREA OF PUNCHING SHEAR

find design punching shear angle (figure 27)

platform material characteristic angle of internal friction:

$$\varphi_{pk} = 40 \text{ deg}$$

Subgrade material characteristic angle of internal friction:

$$\varphi_{sk} = 28 \text{ deg}$$

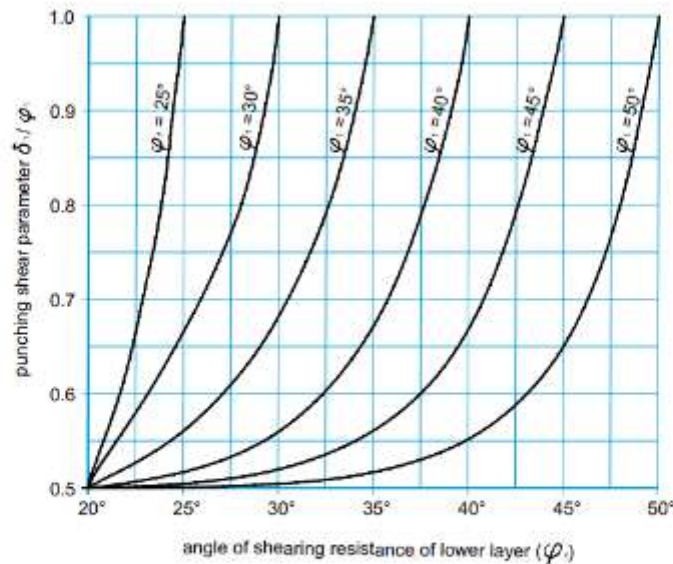


Figure 27 – δ/ϕ for granular formation (after Hanna, 1981)

nominal punching shear parameter (from chart):

$$\rho_{\delta} := \frac{\delta_{pk}}{\varphi_{pk}} \quad \rho_{\delta} := 0.53$$

design punching shear angle:

$$\delta_{pd} := \rho_{\delta} \cdot \varphi_{pd} = 17.95 \text{ deg}$$

5.3 LATERAL LOADS IN PLATFORM

coefficient of active lateral earth pressure for platform:

$$K_{apd} := \left(\frac{\sin(90^{\circ} - \varphi_{pd})}{\sqrt{\sin(90^{\circ} + \delta_{pd})} + \sqrt{(\sin(\varphi_{pd} + \delta_{pd}) \cdot \sin(\varphi_{pd}))}} \right)^2 = 0.26$$

$$K_{ppd} := \left(\frac{\sin(90^{\circ} + \varphi_{pd})}{\sqrt{\sin(90^{\circ} - \delta_{pd})} - \sqrt{(\sin(\varphi_{pd} + \delta_{pd}) \cdot \sin(\varphi_{pd}))}} \right)^2 = 7.02$$

find ratio of applied pressure to effective pressure at mid-depth of platform (figure 25)

mid-point depth of platform

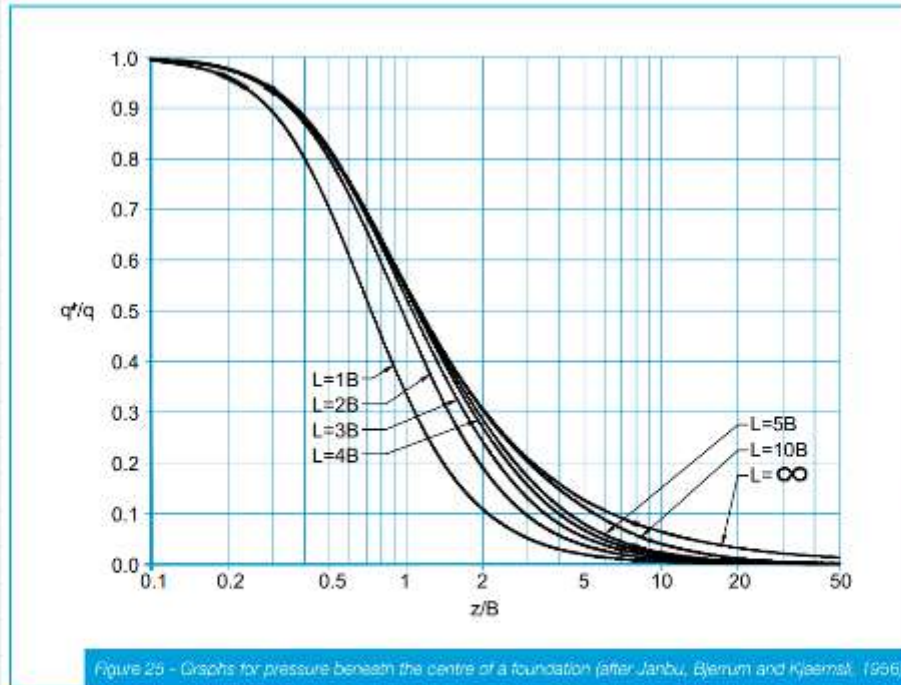
$$D_{mid} := \frac{D}{2} = 0.225 \text{ m}$$

mid-point depth to breadth ratio:

$$\frac{D_{mid}}{B} = 0.45$$

ratio of applied load length to breadth:

$$\frac{L}{B} = 3.9$$



ratio of applied pressure to effective pressure (from chart) $\rho_q := \frac{q_{avd}}{q_d}$ $\rho_q := 0.85$

average vertical pressure due to load: $q_{avd}' := \rho_q \cdot q_d = 158.02 \text{ kPa}$

active lateral load in fill (per lin m): $P_{apd} := K_{apd} \cdot \left(q_{avd}' + \gamma_p \cdot \frac{D}{2} \right) \cdot D = 18.81 \frac{\text{kN}}{\text{m}}$

passive lateral load in fill (per lin m) $P_{ppd} := K_{ppd} \cdot \gamma_p \cdot \frac{D^2}{2} = 14.21 \frac{\text{kN}}{\text{m}}$

5.4 HORIZONTAL AND VERTICAL LOADS ON SUB-GRADE

horizontal load on sub-grade (per lin m) $F_{Hs} := P_{apd} - P_{ppd} = 4.6 \frac{\text{kN}}{\text{m}}$

vertical load on sub-grade (per lin m) $F_{Vs} := \frac{(q_d \cdot B + \gamma_p \cdot D \cdot B')}{2} = 49.94 \frac{\text{kN}}{\text{m}}$

5.5 TOTAL VERTICAL LOAD EFFECT ON SUB-GRADE

characteristic permanent action due to platform self-weight $G_{pk} := \gamma_p \cdot D \cdot B' \cdot L' = 15.39 \text{ kN}$

partial factor for permanent actions: $\gamma_G := 1.0$

design permanent action due to platform self-weight $G_{pd} := \gamma_G \cdot G_{pk} = 15.39 \text{ kN}$

total design vertical action $V_{Ed} := G_{pd} + Q_d = 196.65 \text{ kN}$



5.6 SUB-GRADE BEARING RESISTANCE WITH PLATFORM

bearing capacity factor for gravity term

$$N_{\gamma d} = 5.06$$

Sub-grade material design angle of internal friction (see Section 2):

$$\varphi_{sd} = 23.04 \quad \varphi_{sd} := 23.04^\circ$$

bearing capacity factor for overburden term

$$N_{qd} := e^{\pi \cdot \tan(\varphi_{sd})} \cdot \left(\tan \left(45^\circ + \frac{\varphi_{sd}}{2} \right) \right)^2 = 8.7$$

shape factor for gravity term:

$$s_{\gamma d} := 1 - \left(0.4 \cdot \frac{B'}{L'} \right) = 0.86$$

Shape factor for overburden term:

$$s_{qd} := 1 + \left(\tan(\varphi_{sd}) \cdot \frac{B'}{L'} \right) = 1.15$$

inclination factor exponent:

$$m := \frac{\left(2 + \left(\frac{B'}{L'} \right) \right)}{1 + \left(\frac{B'}{L'} \right)} = 1.74$$

inclination factor for gravity term:

$$i_{\gamma d} := \min \left(1, \left(1 - \frac{F_{Hs}}{F_{Vs}} \right)^{m+1} \right) = 0.77$$

inclination factor for overburden term:

$$i_{qd} := \min \left(1, \left(1 - \frac{F_{Hs}}{F_{Vs}} \right)^m \right) = 0.85$$

depth factor for gravity term;

$$d_{\gamma d} := 1$$

depth factor for overburden term:

$$d_{qd} := 1 + 2 \cdot \tan(\varphi_{sd}) \cdot (1 - \sin(\varphi_{sd}))^2 \cdot \operatorname{atan} \left(\frac{D}{B'} \right) = 1.17$$

groundwater factor for gravity term:

$$w_{\gamma d} := \min \left(1, \max \left(0.5, 0.5 \cdot \left(1 + \frac{z_{\gamma}}{B'} \right) \right) \right) = 1$$

depth of groundwater below top of platform:

$$z_q := z_{\gamma} + D = 1.27 \text{ m}$$

groundwater factor for surcharge term:

$$w_{qd} := \min \left(1, \max \left(0.5, 0.5 \cdot \left(1 + \frac{z_q}{D} \right) \right) \right) = 1$$

bearing resistance (overburden)

$$W_{Rd1} := 0.5 \cdot \gamma_s \cdot B' \cdot N_{\gamma d} \cdot s_{\gamma d} \cdot i_{\gamma d} \cdot d_{\gamma d} \cdot w_{\gamma d} = 23.171 \text{ kPa}$$

bearing resistance (self-weight)

$$W_{Rd2} := \gamma_p \cdot D \cdot N_{qd} \cdot s_{qd} \cdot i_{qd} \cdot d_{qd} \cdot w_{qd} = 88.56 \text{ kPa}$$

total

$$W_{Rd} := W_{Rd1} + W_{Rd2} = 111.731 \text{ kPa}$$

$$L' \cdot B' = 1.711 \text{ m}^2$$

bearing resistance

$$V_{Rd} := W_{Rd} \cdot B' \cdot L' = 191.118 \text{ kN}$$

UTILISATION

$$\frac{V_{Ed}}{V_{Rd}} = 1.029$$

 CASE Consultants Civil And Structural Engineers Moose Hall Barewell Rd Torquay, TQ1 4PA	Project		Noss On Dart Marina, Dartmouth		Job Ref.		C8829
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	Calc. by	Date	Chk'd by	Date	App'd by	Date	
	D.P.H	DEC 23					

SUMMARY – clayey GRAVEL

Proposed thickness of platform = 450mm

Subgrade material characteristic angle of internal friction = 28°

Subgrade material density = 18 kN/m^3

Load Cases:

Load Case	Effective track dimensions and bearing pressure at subgrade formation		
	Length L (m)	Width B (m)	Area A (m^2)
1	2.31	0.5	1.16
2	1.95	0.5	0.98

Load Case	UDL Load (kPa)	Partial Factors γ_Q	Design Applied Pressure q_d (kPa)	Design Load Q_d (kN)
1	123	1.3	159.9	184.7
2	143	1.3	185.9	181.3

Design factors:

Bearing Capacity Factors		Depth factors			Shape factors			Inclination factors		
N	Granular Subgrade	d	Case 1	Case 2	S	Case 1	Case 2	i	Case 1	Case 2
N _q	8.7	d _q	1.17	1.17	S _q	1.13	1.15	i _q	0.92	0.85
N _y	5.06	d _y	1.0	1.0	S _y	0.88	0.86	i _y	0.88	0.77

Design resistances:

Load Case	Overburden (kPa)	Self Weight (kPa)	Total (kPa)	Area (m^2)	VRd (kN)
1	27.1	94.5	121.6	1.99	242.1
2	23.2	88.6	111.7	1.71	191.1

Design actions:

Load Case	Platform Self-weight, G_{pd} (kN)	Variable Load, Q_d (kN)	Total, V_{Ed} (kN)
1	17.9	184.7	202.6
2	15.4	181.3	196.7

UTILISATION:

Load Case	VRd (kN)	V_{Ed} (kN)	Utilization
1	242.1	202.6	84%
2	191.1	196.7	103%

∴ BEARING RESISTANCE OF CLAYEY GRAVEL (MADE GROUND)

IS ADEQUATE WITH 450mm THK PLATFORM



Bovisand Formation

ULS check - LC1 and silty MUDSTONE

1. VARIABLE ACTIONS

applied load breadth:

$$B := 0.5 \text{ m}$$

applied load length:

$$L := 2.31 \text{ m}$$

applied load area:

$$A := B \cdot L = 1.155 \text{ m}^2$$

characteristic applied GBP:

$$q_k := 123 \text{ kPa}$$

Partial factors on variable actions:

$$\gamma_Q := 1.3$$

design applied ground bearing pressure:

$$q_d := \gamma_Q \cdot q_k = 159.9 \text{ kPa}$$

Design applied load:

$$Q_d := q_d \cdot A = 184.7 \text{ kN}$$

2. SUB-GRADE PARAMETERS

Subgrade material characteristic angle of internal friction:

$$\varphi_{sk} := 32^\circ$$

Partial factor on sub-grade strength:

$$\gamma_\varphi := 1.25$$

Sub-grade material design angle of internal friction:

$$\varphi_{sd} := \text{atan} \left(\frac{\tan(\varphi_{sk})}{\gamma_\varphi} \right) = 26.56 \text{ deg}$$

$$\varphi_{sd} := 26.56$$

sub-grade material density:

$$\gamma_s := 24 \frac{\text{kN}}{\text{m}^3}$$

depth of ground water below formation:

$$z_\gamma := 0.82 \text{ m}$$

3. SUB-GRADE BEARING RESISTANCE WITHOUT PLATFORM

Bearing capacity factor for gravity term:

$$N_{\gamma d} := 0.1054 \cdot e^{0.168 \cdot \varphi_{sd}} = 9.13$$

design shape factor for gravity term:

$$s_{\gamma d} := 1 - \left(0.4 \cdot \frac{B}{L} \right) = 0.91$$

design bearing resistance:

$$V_{Rd} := 0.5 \cdot \gamma_s \cdot B \cdot N_{\gamma d} \cdot s_{\gamma d} \cdot A = 57.82 \text{ kN}$$

UTILISATION:

$$\frac{Q_d}{V_{Rd}} = 3.19$$

> 1.0 therefore FAIL &
Platform required



4. PLATFORM PARAMETERS

select platform material based on presumed bearing capacity (table 3):

platform material characteristic angle of internal friction:

$$\varphi_{pk} := 40^\circ$$

partial factor on platform material strength:

$$\gamma_\varphi := 1.25$$

platform material design angle of internal friction:

$$\varphi_{pd} := \operatorname{atan}\left(\frac{\tan(\varphi_{pk})}{\gamma_\varphi}\right) = 33.87 \text{ deg}$$

platform material density:

$$\gamma_p := 20 \frac{\text{kN}}{\text{m}^3}$$

5.1 EFFECTIVE AREA, DIMENSIONS AND LOAD SPREAD ANGLE

PLATFORM THICKNESS:

$$D := 0.45 \text{ m}$$

find ratio of applied pressure to effective pressure at formation (figure 25)

ratio of applied load breadth to platform depth:

$$\frac{D}{B} = 0.9$$

ratio of applied load length to breadth:

$$\frac{L}{B} = 4.62$$

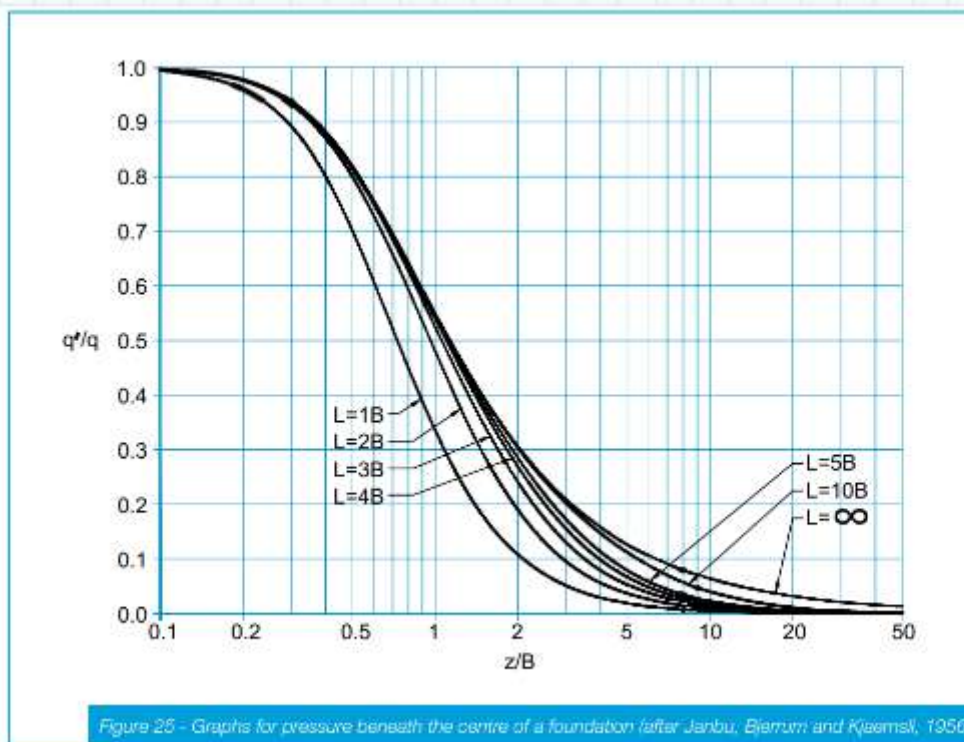


Figure 25 - Graphs for pressure beneath the centre of a foundation (after Janbu, Bjerrum and Kjaemsli, 1956)



ratio of applied pressure to effective pressure
(from chart):

$$\rho_q := \frac{q_d'}{q_d} \quad \rho_q = 0.58$$

design effective pressure:

$$q_d' := \rho_q \cdot q_d = 92.74 \text{ kPa}$$

design effective area:

$$A' := \frac{Q_d}{q_d'} = 1.99 \text{ m}^2$$

final load spread width and angle solving with quadratic equation

quadratic factors:

$$\begin{aligned} a &:= 1 \\ b &:= L + B = 2.81 \text{ m} \\ c &:= A - A' = -0.84 \text{ m}^2 \end{aligned}$$

quadratic solution:

$$x := \frac{\sqrt{(b^2 - 4 \cdot a \cdot c)} - b}{2 \cdot a} = 0.27 \text{ m}$$

load spread width:

$$b' := \frac{x}{2} = 0.14 \text{ m}$$

load spread angle:

$$\beta := \text{atan}\left(\frac{b'}{D}\right) = 16.78 \text{ deg} \quad \beta = 16.78$$

maximum load spread width for $\beta = 26.6$

$$b'_{max} := \frac{D}{2} = 0.225 \text{ m}$$

$b' < b'_{max}$ therefore use b'

effective breadth:

$$B' := \min(B + 2 \cdot b', B + 2 \cdot b'_{max}) = 0.77 \text{ m}$$

effective length:

$$L' := \min(L + 2 \cdot b', L + 2 \cdot b'_{max}) = 2.58 \text{ m}$$

5.2 EFFECTIVE AREA OF PUNCHING SHEAR

find design punching shear angle (figure 27)

platform material characteristic angle of internal friction:

$$\varphi_{pk} = 40 \text{ deg}$$

Subgrade material characteristic angle of internal friction:

$$\varphi_{sk} = 32 \text{ deg}$$

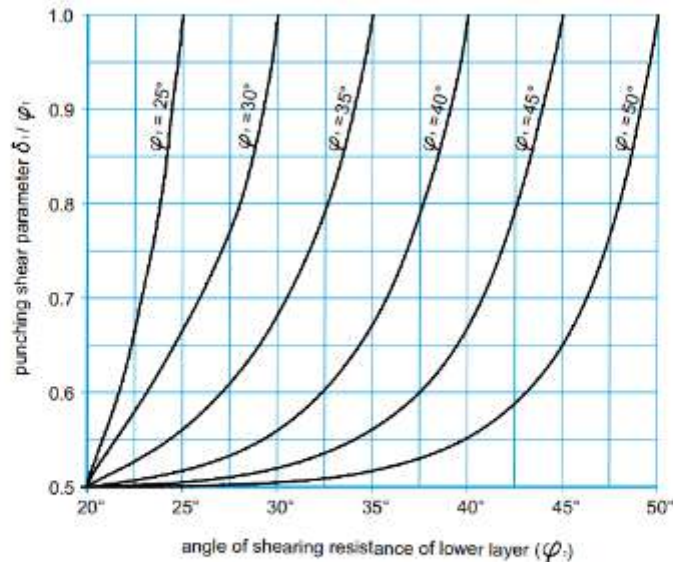


Figure 27 – δ/ϕ for granular formation (after Hanna, 1981)

nominal punching shear parameter (from chart):

$$\rho_{\delta} := \frac{\delta_{pk}}{\varphi_{pk}} \quad \rho_{\delta} := 0.57$$

design punching shear angle:

$$\delta_{pd} := \rho_{\delta} \cdot \varphi_{pd} = 19.31 \text{ deg}$$

5.3 LATERAL LOADS IN PLATFORM

coefficient of active lateral earth pressure for platform:

$$K_{apd} := \left(\frac{\sin(90^\circ - \varphi_{pd})}{\sqrt{\sin(90^\circ + \delta_{pd})} + \sqrt{(\sin(\varphi_{pd} + \delta_{pd}) \cdot \sin(\varphi_{pd}))}} \right)^2 = 0.26$$

$$K_{ppd} := \left(\frac{\sin(90^\circ + \varphi_{pd})}{\sqrt{\sin(90^\circ - \delta_{pd})} - \sqrt{(\sin(\varphi_{pd} + \delta_{pd}) \cdot \sin(\varphi_{pd}))}} \right)^2 = 7.48$$

find ratio of applied pressure to effective pressure at mid-depth of platform (figure 25)

mid-point depth of platform

$$D_{mid} := \frac{D}{2} = 0.225 \text{ m}$$

mid-point depth to breadth ratio:

$$\frac{D_{mid}}{B} = 0.45$$

ratio of applied load length to breadth:

$$\frac{L}{B} = 4.62$$

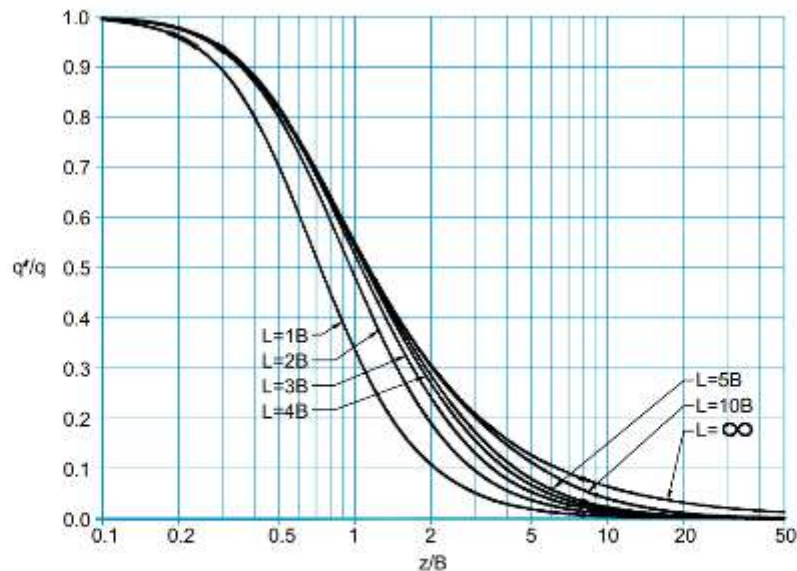


Figure 25 - Graphs for pressure beneath the centre of a foundation (after Janbu, Bjerrum and Kluminski, 1986)

ratio of applied pressure to effective pressure (from chart) $\rho_q := \frac{q_{avd}}{q_d}$ $\rho_q := 0.85$

average vertical pressure due to load:

$$q_{avd}' := \rho_q \cdot q_d = 135.92 \text{ kPa}$$

active lateral load in fill (per lin m):

$$P_{apd} := K_{apd} \cdot \left(q_{avd}' + \gamma_p \cdot \frac{D}{2} \right) \cdot D = 16.21 \frac{\text{kN}}{\text{m}}$$

passive lateral load in fill (per lin m)

$$P_{ppd} := K_{ppd} \cdot \gamma_p \cdot \frac{D^2}{2} = 15.15 \frac{\text{kN}}{\text{m}}$$

5.4 HORIZONTAL AND VERTICAL LOADS ON SUB-GRADE

horizontal load on sub-grade (per lin m)

$$F_{Hs} := P_{apd} - P_{ppd} = 1.05 \frac{\text{kN}}{\text{m}}$$

vertical load on sub-grade (per lin m)

$$F_{Vs} := \frac{(q_d \cdot B + \gamma_p \cdot D \cdot B')}{2} = 43.45 \frac{\text{kN}}{\text{m}}$$

5.5 TOTAL VERTICAL LOAD EFFECT ON SUB-GRADE

characteristic permanent action due to platform self-weight

$$G_{pk} := \gamma_p \cdot D \cdot B' \cdot L' = 17.92 \text{ kN}$$

partial factor for permanent actions:

$$\gamma_G := 1.0$$

design permanent action due to platform self-weight

$$G_{pd} := \gamma_G \cdot G_{pk} = 17.92 \text{ kN}$$

total design vertical action

$$V_{Ed} := G_{pd} + Q_d = 202.61 \text{ kN}$$



5.6 SUB-GRADE BEARING RESISTANCE WITH PLATFORM

bearing capacity factor for gravity term

$$N_{\gamma d} = 9.13$$

Sub-grade material design angle of internal friction (see Section 2):

$$\varphi_{sd} = 26.56 \quad \varphi_{sd} := 26.56^\circ$$

bearing capacity factor for overburden term

$$N_{qd} := e^{\pi \cdot \tan(\varphi_{sd})} \cdot \left(\tan \left(45^\circ + \frac{\varphi_{sd}}{2} \right) \right)^2 = 12.59$$

shape factor for gravity term:

$$s_{\gamma d} := 1 - \left(0.4 \cdot \frac{B'}{L'} \right) = 0.88$$

Shape factor for overburden term:

$$s_{qd} := 1 + \left(\tan(\varphi_{sd}) \cdot \frac{B'}{L'} \right) = 1.15$$

inclination factor exponent:

$$m := \frac{\left(2 + \left(\frac{B'}{L'} \right) \right)}{1 + \left(\frac{B'}{L'} \right)} = 1.77$$

inclination factor for gravity term:

$$i_{\gamma d} := \min \left(1, \left(1 - \frac{F_{Hs}}{F_{Vs}} \right)^{m+1} \right) = 0.93$$

inclination factor for overburden term:

$$i_{qd} := \min \left(1, \left(1 - \frac{F_{Hs}}{F_{Vs}} \right)^m \right) = 0.96$$

depth factor for gravity term;

$$d_{\gamma d} := 1$$

depth factor for overburden term:

$$d_{qd} := 1 + 2 \cdot \tan(\varphi_{sd}) \cdot (1 - \sin(\varphi_{sd}))^2 \cdot \operatorname{atan} \left(\frac{D}{B'} \right) = 1.16$$

groundwater factor for gravity term:

$$w_{\gamma d} := \min \left(1, \max \left(0.5, 0.5 \cdot \left(1 + \frac{z_{\gamma}}{B'} \right) \right) \right) = 1$$

depth of groundwater below top of platform:

$$z_q := z_{\gamma} + D = 1.27 \text{ m}$$

groundwater factor for surcharge term:

$$w_{qd} := \min \left(1, \max \left(0.5, 0.5 \cdot \left(1 + \frac{z_q}{D} \right) \right) \right) = 1$$

bearing resistance (overburden)

$$W_{Rd1} := 0.5 \cdot \gamma_s \cdot B' \cdot N_{\gamma d} \cdot s_{\gamma d} \cdot i_{\gamma d} \cdot d_{\gamma d} \cdot w_{\gamma d} = 69.563 \text{ kPa}$$

bearing resistance (self-weight)

$$W_{Rd2} := \gamma_p \cdot D \cdot N_{qd} \cdot s_{qd} \cdot i_{qd} \cdot d_{qd} \cdot w_{qd} = 144.796 \text{ kPa}$$

total

$$W_{Rd} := W_{Rd1} + W_{Rd2} = 214.359 \text{ kPa}$$

$$L' \cdot B' = 1.991 \text{ m}^2$$

bearing resistance

$$V_{Rd} := W_{Rd} \cdot B' \cdot L' = 426.87 \text{ kN}$$

UTILISATION

$$\frac{V_{Ed}}{V_{Rd}} = 0.475$$



ULS check - LC2 and silty MUDSTONE

1. VARIABLE ACTIONS

applied load breadth:

$$B := 0.5 \text{ m}$$

applied load length:

$$L := 1.95 \text{ m}$$

applied load area:

$$A := B \cdot L = 0.975 \text{ m}^2$$

characteristic applied GBP:

$$q_k := 143 \text{ kPa}$$

Partial factors on variable actions:

$$\gamma_Q := 1.3$$

design applied ground bearing pressure:

$$q_d := \gamma_Q \cdot q_k = 185.9 \text{ kPa}$$

Design applied load:

$$Q_d := q_d \cdot A = 181.3 \text{ kN}$$

2. SUB-GRADE PARAMETERS

Subgrade material characteristic angle of internal friction:

$$\varphi_{sk} := 32^\circ$$

Partial factor on sub-grade strength:

$$\gamma_\varphi := 1.25$$

Sub-grade material design angle of internal friction:

$$\varphi_{sd} := \text{atan} \left(\frac{\tan(\varphi_{sk})}{\gamma_\varphi} \right) = 26.56 \text{ deg}$$

$$\varphi_{sd} := 26.56$$

sub-grade material density:

$$\gamma_s := 24 \frac{\text{kN}}{\text{m}^3}$$

depth of ground water below formation:

$$z_\gamma := 0.82 \text{ m}$$

3. SUB-GRADE BEARING RESISTANCE WITHOUT PLATFORM

Bearing capacity factor for gravity term:

$$N_{\gamma d} := 0.1054 \cdot e^{0.168 \cdot \varphi_{sd}} = 9.13$$

design shape factor for gravity term:

$$s_{\gamma d} := 1 - \left(0.4 \cdot \frac{B}{L} \right) = 0.9$$

design bearing resistance:

$$V_{Rd} := 0.5 \cdot \gamma_s \cdot B \cdot N_{\gamma d} \cdot s_{\gamma d} \cdot A = 47.96 \text{ kN}$$

UTILISATION:

$$\frac{Q_d}{V_{Rd}} = 3.78$$

> 1.0 therefore FAIL &
Platform required



4. PLATFORM PARAMETERS

select platform material based on presumed bearing capacity (table 3):

platform material characteristic angle of internal friction:

$$\varphi_{pk} := 40^\circ$$

partial factor on platform material strength:

$$\gamma_\varphi := 1.25$$

platform material design angle of internal friction:

$$\varphi_{pd} := \text{atan} \left(\frac{\tan(\varphi_{pk})}{\gamma_\varphi} \right) = 33.87 \text{ deg}$$

platform material density:

$$\gamma_p := 20 \frac{\text{kN}}{\text{m}^3}$$

5.1 EFFECTIVE AREA, DIMENSIONS AND LOAD SPREAD ANGLE

PLATFORM THICKNESS:

$$D := 0.45 \text{ m}$$

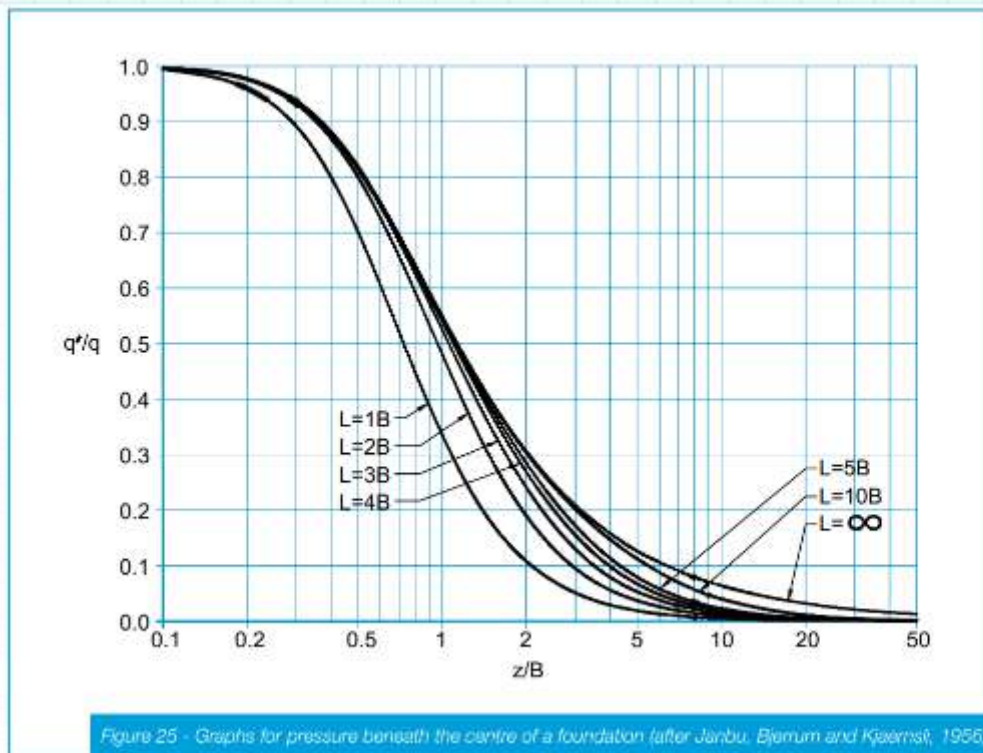
find ratio of applied pressure to effective pressure at formation (figure 25)

ratio of applied load breadth to platform depth:

$$\frac{D}{B} = 0.9$$

ratio of applied load length to breadth:

$$\frac{L}{B} = 3.9$$





ratio of applied pressure to effective pressure
(from chart):

$$\rho_q := \frac{q_d'}{q_d} \quad \rho_q := 0.57$$

design effective pressure:

$$q_d' := \rho_q \cdot q_d = 105.96 \text{ kPa}$$

design effective area:

$$A' := \frac{Q_d}{q_d'} = 1.71 \text{ m}^2$$

final load spread width and angle solving with quadratic equation

quadratic factors:

$$\begin{aligned} a &:= 1 \\ b &:= L + B = 2.45 \text{ m} \\ c &:= A - A' = -0.74 \text{ m}^2 \end{aligned}$$

quadratic solution:

$$x := \frac{\sqrt{(b^2 - 4 \cdot a \cdot c)} - b}{2 \cdot a} = 0.27 \text{ m}$$

load spread width:

$$b' := \frac{x}{2} = 0.14 \text{ m}$$

load spread angle:

$$\beta := \text{atan}\left(\frac{b'}{D}\right) = 16.72 \text{ deg}$$

$$\beta := 16.72$$

maximum load spread width for $\beta = 26.6$

$$b'_{max} := \frac{D}{2} = 0.225 \text{ m}$$

$b' < b'_{max}$ therefore use b'

effective breadth:

$$B' := \min(B + 2 \cdot b', B + 2 \cdot b'_{max}) = 0.77 \text{ m}$$

effective length:

$$L' := \min(L + 2 \cdot b', L + 2 \cdot b'_{max}) = 2.22 \text{ m}$$

5.2 EFFECTIVE AREA OF PUNCHING SHEAR

find design punching shear angle (figure 27)

platform material characteristic angle of internal friction:

$$\varphi_{pk} = 40 \text{ deg}$$

Subgrade material characteristic angle of internal friction:

$$\varphi_{sk} = 32 \text{ deg}$$

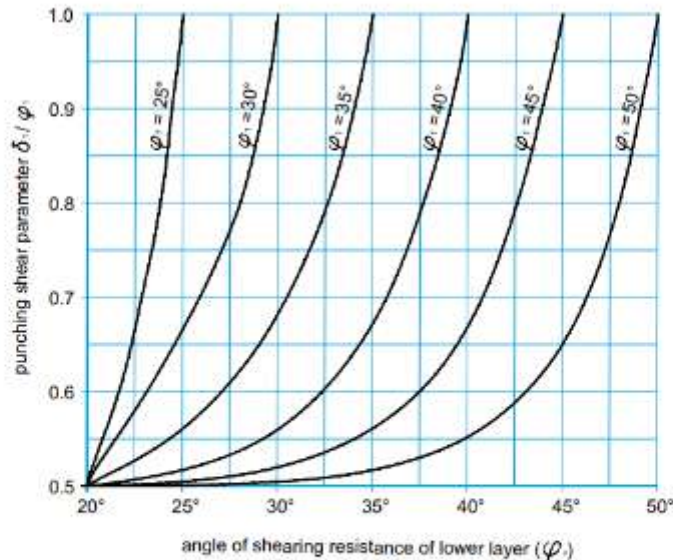


Figure 27 – δ/ϕ for granular formation (after Hanna, 1981)

nominal punching shear parameter (from chart):

$$\rho_{\delta} := \frac{\delta_{pk}}{\varphi_{pk}} \quad \rho_{\delta} := 0.57$$

design punching shear angle:

$$\delta_{pd} := \rho_{\delta} \cdot \varphi_{pd} = 19.31 \text{ deg}$$

5.3 LATERAL LOADS IN PLATFORM

coefficient of active lateral earth pressure for platform:

$$K_{apd} := \left(\frac{\sin(90^\circ - \varphi_{pd})}{\sqrt{\sin(90^\circ + \delta_{pd})} + \sqrt{(\sin(\varphi_{pd} + \delta_{pd}) \cdot \sin(\varphi_{pd}))}} \right)^2 = 0.26$$

$$K_{ppd} := \left(\frac{\sin(90^\circ + \varphi_{pd})}{\sqrt{\sin(90^\circ - \delta_{pd})} - \sqrt{(\sin(\varphi_{pd} + \delta_{pd}) \cdot \sin(\varphi_{pd}))}} \right)^2 = 7.48$$

find ratio of applied pressure to effective pressure at mid-depth of platform (figure 25)

mid-point depth of platform

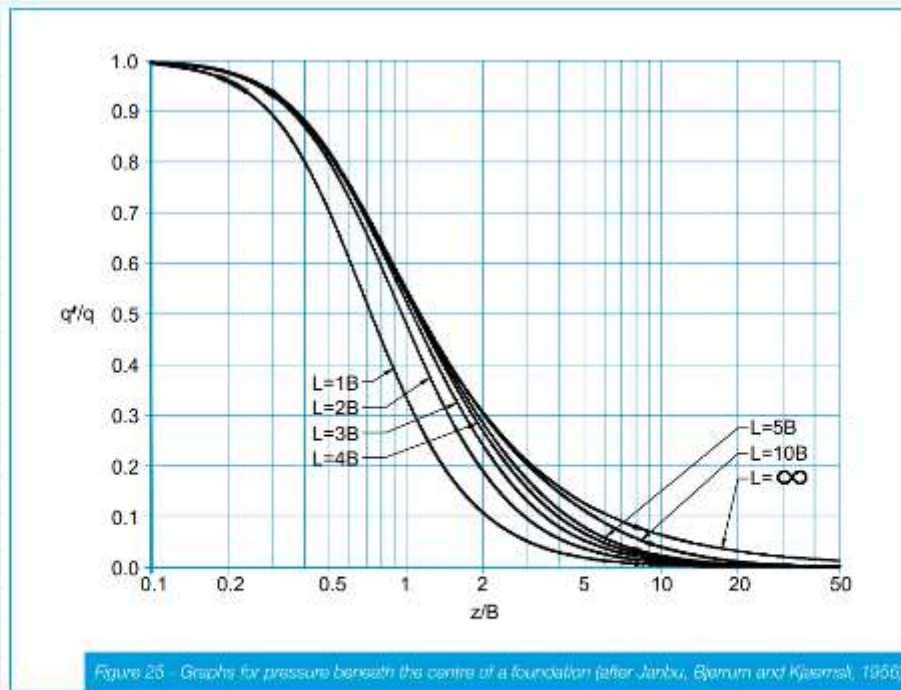
$$D_{mid} := \frac{D}{2} = 0.225 \text{ m}$$

mid-point depth to breadth ratio:

$$\frac{D_{mid}}{B} = 0.45$$

ratio of applied load length to breadth:

$$\frac{L}{B} = 3.9$$



ratio of applied pressure to effective pressure (from chart) $\rho_q := \frac{q_{avd}}{q_d}$ $\rho_q := 0.85$

average vertical pressure due to load:

$$q_{avd}' := \rho_q \cdot q_d = 158.02 \text{ kPa}$$

active lateral load in fill (per lin m):

$$P_{apd} := K_{apd} \cdot \left(q_{avd}' + \gamma_p \cdot \frac{D}{2} \right) \cdot D = 18.76 \frac{\text{kN}}{\text{m}}$$

passive lateral load in fill (per lin m)

$$P_{ppd} := K_{ppd} \cdot \gamma_p \cdot \frac{D^2}{2} = 15.15 \frac{\text{kN}}{\text{m}}$$

5.4 HORIZONTAL AND VERTICAL LOADS ON SUB-GRADE

horizontal load on sub-grade (per lin m)

$$F_{Hs} := P_{apd} - P_{ppd} = 3.6 \frac{\text{kN}}{\text{m}}$$

vertical load on sub-grade (per lin m)

$$F_{Vs} := \frac{(q_d \cdot B + \gamma_p \cdot D \cdot B)}{2} = 49.94 \frac{\text{kN}}{\text{m}}$$

5.5 TOTAL VERTICAL LOAD EFFECT ON SUB-GRADE

characteristic permanent action due to platform self-weight $G_{pk} := \gamma_p \cdot D \cdot B' \cdot L' = 15.39 \text{ kN}$

partial factor for permanent actions:

$$\gamma_G := 1.0$$

design permanent action due to platform self-weight

$$G_{pd} := \gamma_G \cdot G_{pk} = 15.39 \text{ kN}$$

total design vertical action

$$V_{Ed} := G_{pd} + Q_d = 196.65 \text{ kN}$$



5.6 SUB-GRADE BEARING RESISTANCE WITH PLATFORM

bearing capacity factor for gravity term

$$N_{\gamma d} = 9.13$$

Sub-grade material design angle of internal friction (see Section 2):

$$\varphi_{sd} = 26.56 \quad \varphi_{sd} := 26.56^\circ$$

bearing capacity factor for overburden term

$$N_{qd} := e^{\pi \cdot \tan(\varphi_{sd})} \cdot \left(\tan\left(45^\circ + \frac{\varphi_{sd}}{2}\right) \right)^2 = 12.59$$

shape factor for gravity term:

$$s_{\gamma d} := 1 - \left(0.4 \cdot \frac{B'}{L'} \right) = 0.86$$

Shape factor for overburden term:

$$s_{qd} := 1 + \left(\tan(\varphi_{sd}) \cdot \frac{B'}{L'} \right) = 1.17$$

inclination factor exponent:

$$m := \frac{\left(2 + \left(\frac{B'}{L'} \right) \right)}{1 + \left(\frac{B'}{L'} \right)} = 1.74$$

inclination factor for gravity term:

$$i_{\gamma d} := \min \left(1, \left(1 - \frac{F_{Hs}}{F_{Vs}} \right)^{m+1} \right) = 0.81$$

inclination factor for overburden term:

$$i_{qd} := \min \left(1, \left(1 - \frac{F_{Hs}}{F_{Vs}} \right)^m \right) = 0.88$$

depth factor for gravity term;

$$d_{\gamma d} := 1$$

depth factor for overburden term:

$$d_{qd} := 1 + 2 \cdot \tan(\varphi_{sd}) \cdot (1 - \sin(\varphi_{sd}))^2 \cdot \operatorname{atan}\left(\frac{D}{B'}\right) = 1.16$$

groundwater factor for gravity term:

$$w_{\gamma d} := \min \left(1, \max \left(0.5, 0.5 \cdot \left(1 + \frac{z_{\gamma}}{B'} \right) \right) \right) = 1$$

depth of groundwater below top of platform:

$$z_q := z_{\gamma} + D = 1.27 \text{ m}$$

groundwater factor for surcharge term:

$$w_{qd} := \min \left(1, \max \left(0.5, 0.5 \cdot \left(1 + \frac{z_q}{D} \right) \right) \right) = 1$$

bearing resistance (overburden)

$$W_{Rd1} := 0.5 \cdot \gamma_s \cdot B' \cdot N_{\gamma d} \cdot s_{\gamma d} \cdot i_{\gamma d} \cdot d_{\gamma d} \cdot w_{\gamma d} = 59.225 \text{ kPa}$$

bearing resistance (self-weight)

$$W_{Rd2} := \gamma_p \cdot D \cdot N_{qd} \cdot s_{qd} \cdot i_{qd} \cdot d_{qd} \cdot w_{qd} = 135.52 \text{ kPa}$$

total

$$W_{Rd} := W_{Rd1} + W_{Rd2} = 194.745 \text{ kPa}$$

$$L' \cdot B' = 1.711 \text{ m}^2$$

bearing resistance

$$V_{Rd} := W_{Rd} \cdot B' \cdot L' = 333.116 \text{ kN}$$

UTILISATION

$$\frac{V_{Ed}}{V_{Rd}} = 0.59$$

 CASE Consultants Civil And Structural Engineers Moose Hall Barewell Rd Torquay, TQ1 4PA	Project		Noss On Dart Marina, Dartmouth		Job Ref.		C8829	
	Section		GEAX EK60L Piling Rig		Sheet no./rev.		G 27	
	Calc. by	Date	Chk'd by	Date	App'd by	Date		
	D.P.H	DEC 23						

SUMMARY – slaty MUDSTONE

Proposed thickness of platform = 450mm

Subgrade material characteristic angle of internal friction = 32°

Subgrade material density = 24 kN/m^3

Load Cases:

Load Case	Effective track dimensions and bearing pressure at subgrade formation		
	Length L (m)	Width B (m)	Area A (m^2)
1	2.31	0.5	1.16
2	1.95	0.5	0.98

Load Case	UDL Load (kPa)	Partial Factors γ_Q	Design Applied Pressure q_d (kPa)	Design Load Q_d (kN)
1	123	1.3	159.9	184.7
2	143	1.3	261.3	181.3

Design factors:

Bearing Capacity Factors		Depth factors			Shape factors			Inclination factors		
N	Granular Subgrade	d	Case 1	Case 2	S	Case 1	Case 2	i	Case 1	Case 2
N _q	12.59	d _q	1.16	1.16	S _q	1.15	1.17	i _q	0.96	0.88
N _γ	9.13	d _γ	1.0	1.0	S _γ	0.88	0.86	i _γ	0.93	0.81

Design resistances:

Load Case	Overburden (kPa)	Self Weight (kPa)	Total (kPa)	Area (m^2)	VR _d (kN)
1	77.2	169.9	247.2	2.1	519.0
2	69.6	144.8	214.4	1.99	426.9

Design actions:

Load Case	Platform Self-weight, G _p (kN)	Variable Load, Q _d (kN)	Total, V _{Ed} (kN)
1	21	184.7	205.7
2	17.9	181.3	202.6

UTILISATION:

Load Case	VR _d (kN)	V _{Ed} (kN)	Utilization
1	519.0	205.7	40%
2	426.9	202.6	48%

∴ BEARING RESISTANCE OF SLATY MUDSTONE (BOVISAND FORMATION) IS ADEQUATE WITH 450mm THK PLATFORM



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Project

Noss On Dart Marina, Dartmouth

Job Ref.

C8829

Section

GEAX EK60L Piling Rig

Sheet no./rev.

G 28

Calc. by

D.P.H

Date

DEC 23

Chk'd by

Date

App'd by

Date

Tensor Analysis

LC1 and clayey GRAVEL (Made ground)

Loading

Pressure distribution ②

Rectangle

Effective loaded length, L_{eq}

②

2.31

m

Loaded width, W ②

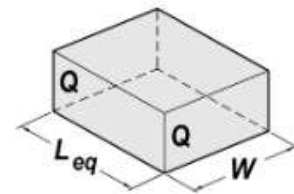
0.5

m

Pressure, Q ②

123

kPa



Platform ②

Material ②

Well-graded crushed rock

Bulk unit weight ②

20

kN/m³

Friction angle ②

40

deg

Max. particle size ②

75

mm

Bearing capacity factor of safety ②

2

Subgrade ②

Subgrade type ②

Granular

Bulk unit weight

18

kN/m³

Friction angle

28

deg

Groundwater

Condition

At or below existing grade

Depth ②

0.8

m

Stabilised ②
US\$23.19/m²

Non-stabilised
\$50.65/m²



	Stabilised	Non-stabilised
Total platform thickness (mm)	300	850
Tensor geogrid	TF055B ②	
Effective loaded length, L_{eq} (m) ②	2.31	2.31
Loaded width (m)	0.5	0.5
Ultimate bearing capacity, q_u (kPa)	362	248
Applied pressure on platform, Q (kPa) ②	123	123
Bearing capacity factor of safety	2.94	2.02
Edge distance, L_e (m) ②	≥ 2	



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Date

LC2 and clayey GRAVEL (Made ground)

Loading

Pressure distribution?

Rectangle

Effective loaded length, L_{eq}

?

1.95

m

Loaded width, W ?

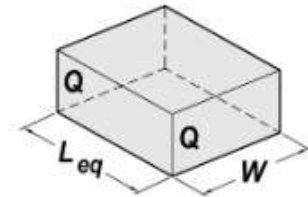
0.5

m

Pressure, Q ?

143

kPa



Platform?

Material?

Well-graded crushed rock

Bulk unit weight?

20

kN/m³

Friction angle?

40

deg

Max. particle size?

75

mm

Bearing capacity factor of safety?

2

Subgrade?

Subgrade type?

Granular

Bulk unit weight

18

kN/m³

Friction angle

28

deg

Groundwater

Condition

At or below existing grade

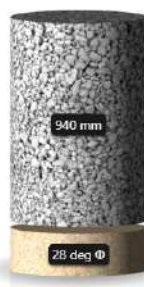
Depth?

0.8

m

Stabilised?
US\$23.19/m²

Non-stabilised
\$56.01/m²



	Stabilised	Non-stabilised
Total platform thickness (mm)	300	940
Tensar geogrid	TF055B?	
Effective loaded length, L_{eq} (m)?	1.95	1.95
Loaded width (m)	0.5	0.5
Ultimate bearing capacity, q_u (kPa)	378	288
Applied pressure on platform, Q (kPa)?	143	143
Bearing capacity factor of safety	2.65	2.02
Edge distance, L_e (m)?	≥ 2	



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LC1 and silty MUDSTONE (Bovisand Formation)

Loading

Pressure distribution ?

Rectangle

Effective loaded length, L_{eq}

?

2.31

m

Loaded width, W ?

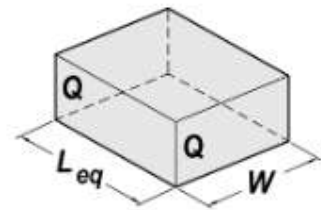
0.5

m

Pressure, Q ?

123

kPa



Platform ?

Material ?

Well-graded crushed rock

Bulk unit weight ?

20

kN/m³

Friction angle ?

40

deg

Max. particle size ?

75

mm

Bearing capacity factor of safety ?

2

Subgrade ?

Subgrade type ?

Granular

Bulk unit weight

24

kN/m³

Friction angle

32

deg

Groundwater

Condition

At or below existing grade

Depth ?

0.8

m

Stabilised ?

US\$23.19/m²

Non-stabilised

\$37.54/m²



	Stabilised	Non-stabilised
Total platform thickness (mm)	300	630
Tensor geogrid	TF055B ?	
Effective loaded length, L_{eq} (m) ?	2.31	2.31
Loaded width (m)	0.5	0.5
Ultimate bearing capacity, q_u (kPa)	560	248
Applied pressure on platform, Q (kPa) ?	123	123
Bearing capacity factor of safety	4.55	2.02
Edge distance, L_e (m) ?	≥ 2	



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LC2 and silty MUDSTONE (Bovisand Formation)

Loading

Pressure distribution ?

Rectangle

Effective loaded length, L_{eq}

?

1.95

m

Loaded width, W ?

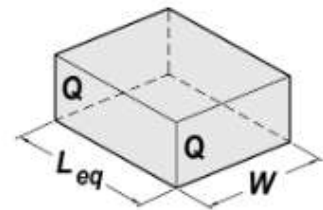
0.5

m

Pressure, Q ?

143

kPa



Platform ?

Material ?

Well-graded crushed rock

Bulk unit weight ?

20

kN/m³

Friction angle ?

40

deg

Max. particle size ?

75

mm

Bearing capacity factor of safety ?

2

Subgrade ?

Subgrade type ?

Granular

Bulk unit weight

24

kN/m³

Friction angle

32

deg

Groundwater

Condition

At or below existing grade

Depth ?

0.8

m

Stabilised ?
US\$23.19/m²

Non-stabilised
\$44.69/m²



	Stabilised	Non-stabilised
Total platform thickness (mm)	300	750
Tensar geogrid	TF055B ?	
Effective loaded length, L_{eq} (m) ?	1.95	1.95
Loaded width (m)	0.5	0.5
Ultimate bearing capacity, q_u (kPa)	584	288
Applied pressure on platform, Q (kPa) ?	143	143
Bearing capacity factor of safety	4.08	2.02
Edge distance, L_e (m) ?	≥ 2	

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Conclusion

1. Piling platform to be 450mm of 6F5 material in accordance with table 6 / 2 of the Specification for Highway Works. To be compacted in layers in accordance with table 6 / 4 of Specification for Highway Works.
2. As an alternative use 300mm of 6F5 material with Tensar InterAx NX850 base geogrid.
3. Following completion of the installation of the mat the surface should be tested using 450mm diameter plate bearing tests to a minimum load of 400kN/m². The maximum residual settlement following this load should not exceed 5.0mm. Minimum of 3 No. tests at the rate of 1 test / 1000m² .